

Sourav Kumar Agarwal

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SKILLS

- **Languages:** JavaScript, TypeScript, SQL
- **Frontend:** React.js, Next.js, Redux Toolkit, HTML5, CSS3, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, Microservices
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **Messaging:** Apache Kafka
- **Tools:** Docker, Git, GitHub, Postman

EDUCATION

- **Rajasthan Technical University** Kota, India
Bachelor of Technology in Information Technology; GPA: 9.07 Aug 2021 – May 2025
- **Cotton University** Guwahati, India
Higher Secondary Education (Class XII); Percentage: 66.80% Aug 2019 – May 2020

EXPERIENCE

- **Next24tech Technology and Services** Remote
Full Stack Developer Intern May 2024 – July 2024
 - **Frontend Development:** Developed responsive web applications using React.js and Next.js across desktop and mobile platforms.
 - **Performance Optimization:** Improved frontend performance by approximately 30% through code splitting, lazy loading, and optimization techniques.
 - **Reusable Component Architecture:** Built reusable UI components and integrated REST APIs for scalable frontend architecture.
 - **Authentication & Security:** Implemented authentication workflows, route protection, and secure access control mechanisms.
 - **Team Collaboration:** Collaborated with cross-functional teams using Git workflows and Agile methodologies.

PROJECTS

- **Microservices E-Commerce Platform** GitHub
Next.js, Node.js, PostgreSQL, Redis, Kafka, Docker, Redis, Turborepo
 - **Microservices Architecture:** Designed independent services for authentication, products, orders, payments, and notifications using a scalable microservices architecture.
 - **Event-Driven Communication:** Implemented Apache Kafka for asynchronous communication between services, ensuring reliable event processing and decoupled workflows.
 - **Caching Layer:** Integrated Redis caching to reduce database load and improve API response times for frequently accessed resources.
 - **Frontend Development:** Built responsive and accessible user interfaces using Next.js, React, and Tailwind CSS with server-side rendering support.
- **AI-Powered Document Chat (RAG)** GitHub
Next.js, Node.js, LangChain, Vector Database, Gemini API
 - **Retrieval-Augmented Generation:** Developed a document question-answering system using RAG pipelines with semantic search and contextual retrieval.
 - **Document Processing:** Implemented document chunking, embedding generation, vector indexing, and similarity search for efficient information retrieval.
 - **LLM Integration:** Integrated LangChain and Gemini API to generate context-aware responses grounded in uploaded PDF documents.
 - **User Experience:** Built responsive chat interfaces, document management workflows, and real-time interactions using Next.js and React.